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RESEARCH ARTICLE

Bi-Modal Multiperspective Percussive (BiMP) Dataset for Visual and Audio Human Fall Detection

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ABSTRACT Falls are the leading cause of injuries and fatalities among older individuals propagating concerns for health and safety. Mitigating these concerns requires timely intervention and response to minimize health complications. Autonomous fall detection systems are often offered as a means to alleviate these concerns. Typical fall detection systems classify a fall event using either inertial- or vision-based data. Aside from these two modes of input for fall detection, sparse human fall audio data is available. Audio datasets associated with the dynamic impact of human falls from the perspective of a spatial environment are both unavailable publicly and focused on niche applications. This dataset provides multimodal composition of visual and audio data to aid in fall detection technique development. Though this dataset emphasizes audio novelty, a synchronized visual component is included offering a multifaceted collection of data. Audio input for fall detection systems presents an opportunity for new approaches towards autonomous fall detection. This dataset comprises human movement representing activities of daily living and falls of 25 participants in various residential settings, yielding 1,300 instances of unique visual and audio samples. Promising utility of the bi-modal multiperspective percussive (BiMP) dataset is demonstrated through experimental data evaluations using techniques including: GoogLeNet, Long Short Term Memory, Continuous Wavelet Transforms, and Short-time Fourier Transforms for human fall detection achieving accuracies up to 96%. The concept of audio-based fall detection has the potential to mitigate concerns regarding privacy and invasiveness, in addition to broadening the scope of fall detection mechanisms.

INDEX TERMS Assisted living, audio analysis, fall detection, machine learning, multimodal dataset.

I. INTRODUCTION

Falls in older adults are a significant public health issue with profound effects on independence, quality of life, and overall healthcare costs [1]. Falls being the leading cause of injuries and injury-related deaths among older adults [2], [3], increases demand for effective fall detection and prevention (FDP) support mechanisms. Physical impacts of falling cause complications for older adults who wish to

remain living in their homes. In addition to potential physical complications, psychological effects including anxiety towards falling again, and reduced levels of activity and social engagement exacerbate risks of falling and diminish the overall quality of life [4], [5]. Aside from the physical and mental health effects of falls, frequency and volume of falls of older individuals are projected to increase, giving rise to additional economic motivations for mitigation of the older adult fall epidemic [6]. Consequently, the development and implementation of effective FDP systems are crucial for supporting the aging population, ensuring their safety,

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and maintaining a sustainable quality of life. The common theme motivating development of FDP schemes revolves around an effort to minimize the severe outcomes and adverse effects stemming from falls impacting the older adult demographic [7].

From a high-level perspective, most FDP systems share a general mode of operation, involving input data from real-time monitoring and processing the data through a classifier. Variations of FDP schemes differ in aspects such as architecture, implementation, mode of execution, and data format being generalized into two categories on the basis visual or sensor data [8]. The two categories encompass a broad variety of visual and sensor data arranged in diverse formats, and have been instrumental towards establishing FDP mechanisms.

Aside from visual- and sensor-based approaches towards FDP systems development, with exceptions such as [9] and [10], the potential for audio fall detection has been largely overlooked. Both existing audio-based FDPs and the utilities necessary for their development are scarce. Notwithstanding the resource of already existent and not publicly available datasets, i.e., [9] and [10], audio data from falls is not commonly recognized for or preserved as a viable means of input for fall detection mechanisms. Characterisation of falls in terms of audio information has the potential to invoke more avenues of autonomous fall detection.

Audio fall detection, as presented in scenarios such as [9], provide visual-audio recordings inclusive of verbal responses of older adults during falls. Similarly, [10] perpetuates falling audio for: verbal response, body collision, and background noise using wearable cameras for data collection. These efforts provide a benchmark for application of audio fall detection, establishing nuance within the FDP context. Furthering audio fall detection, this work aims to provide fall audio independent of verbal utterance from an environmental perspective. The focus is to capture the dynamic impact audio of a fall from a listening device in proximity of a person, as opposed to being attached. Although verbal interference from a fallen person merits consideration, so too do fall instances when no verbal response is able to be provided by a fallen individual.

In this paper, a multimodal and multiperspective human fall dataset with synchronized audio and visual formats from a spatial environmental perspective is developed and evaluated using deep learning and data processing techniques. This dataset places emphasis on the collection and inclusion of dynamic impact fall audio. Further, this dataset aims to address the lack of publicly available audio fall impact data independent of verbal input from an environmental perspective, capturing the audio essence of a human fall as accurately as possible. Recognizing the utility of video data in the context of autonomous fall detection, the audio data presented is accompanied by complimentary video data. Inclusion of both audio and visual data offers the capacity for multimodal analysis and evaluation of fall detection methods, opposed to analyses using a singular visual- or audio-

input. To evaluate the developed bi-modal multiperspective percussive (BiMP) dataset, this paper devises multiple FDP methods, using deep learning as well as image and audio processing techniques, for fall classification. The devised FDP methods not only demonstrate the potential of the BiMP dataset for FDP system development and evaluation, but also prove to be effective FDP methods for fall detection and classification.

Fall detection systems can be trained using this dataset and implemented in environments with expected older adult ambulation. This dataset provides a means to expand the utility of smart home and mobile robotic devices, thus promoting safety for older adults. For example, fall detection systems can be integrated into one or more of the many smart home devices that are already present in the homes of older adults, which capture audio or visual data.

This paper includes a review of work related to visual, sensorial, and audio fall detection mechanisms, along with a review of available human fall datasets (Section II), followed by the methodology of this dataset's construction (Section III). The methodology discusses the breakdown of participants, environments, equipment used, and fall characteristics exhibited in this dataset. It also includes the demographic statistics and physical characteristics of participants, environmental considerations, and fall simulation instructions. Furthermore, the storage and processing of data is discussed to articulate the hierarchy of data storage, along with precautions taken to ensure participant privacy (Section IV). Additional analysis (Section V), results (Section VI), limitations (Section VI-B), and conclusions (Section VII) are discussed providing context towards the viability of this dataset for FDP system development and evaluation. This work was conducted under Clarkson University's Institutional Review Board (IRB) approval number 24-06.

II. RELATED WORK

The following sections are intended to articulate some relevant FDP systems and the methods on which they rely. The first section discusses available FDPs that rely on video input and the techniques leading to fall classification. Similarly, the following section discusses FDPs reliant on sensor-based data, and the methods for classification used. The section following discusses methods used to classify a fall based on audio input, while the final section offers insights on already available fall datasets.

A. VISUAL-BASED FDP SYSTEMS

Fall detection using visual inputs implement various techniques of image processing. Conventional practices in the analysis of visual data can be typically achieved through image manipulation or using key point locations, examples of which can be found in [11]. Examples of image manipulation would include masking, augmentation, or background extraction. Alternatively, methods using key point locations typically define a systematic classification of joint locations or features. Analyses using key points typically result in

some rendition of a skeleton re-composition of extracted joint position information. Image manipulation or key point location are commonly implemented though are not criteria for visual methods. Additionally these modes of analysis can be used in parallel or in tandem with each other.

Image manipulation can be seen in the method presented in [12] to detect falls using human body shape. Exemplifying the usage of background removal, this technique extracted background information from each video frame and used the leftover foreground information in the determination of a fall. Similarly, background segmentation in [13] was accomplished through converting red-green-blue (RGB) inputs to binary images using a mask region-based convolutional neural network (R-CNN). This study used both a publicly available dataset University of Rzeszów (UR) fall detection dataset [14] in conjunction with a self-collected dataset. Fall detection was achieved using secondary convolutional neural network (CNN) feature extractions as inputs of an attention guided bi-directional long short-term memory (LSTM) classifier. Using a specified human posture model derived from joint configurations, [15] extrapolated kinetic information of a person in an RGB video as input to a logistic regression classifier for fall determination. Additionally, [15] constructed a custom dataset in combination with the FallVideo dataset from [16]. Further examples of visual fall detection were presented in [17] using image manipulation and key joint locations.

Despite the application of work done previously, limitations include classification bias due to low variation of data, and requirements for continuous visual monitoring. Nonetheless, available work involving visual inputs for fall detection demonstrate the utility and application of visual fall datasets. Additional to the datasets used to develop existing visual fall detection techniques, a higher volume of visual fall data, from alternative viewpoints offers a more diverse sample set on which to train fall detection mechanisms. Fall data from multiple perspectives effectively facilitates a wider range of view angles from which a fall can be detected.

B. SENSOR-BASED FDP SYSTEMS

Sensor-based fall detection involves a vast selection of sensors. Some sensors are intended to be worn, and others are intended to be integrated into an environment, such as floors. These include, but are not limited to, sensors such as: pressure plates, accelerometers, gyroscopes, manometers, and inertial measurement units. Sensor-based fall detection can be accomplished using a single type of sensor, or a combination of multiple sensors in a multi-input mode of detection.

For instance, [18] utilized data from a smartphone accelerometer in combination with a threshold-based method to detect different types of events, e.g., a fall-like event, such as lying on the floor, or a sudden stop after walking. If a fall-like event was detected by the threshold-based method, a multiple kernel learning support vector machine

was implemented to confirm a fall, or a nonfall. Similarly, an inertial measurement unit combining an accelerometer, gyroscope, and magnetometer were used in [19]. Using the data from these devices, an evaluation was then carried out for classifications using: a threshold based method, a threshold based method with Madgwick's Decomposition, and machine learning methods. Using a piezoelectric sensor, [20] developed a fall detection system based on floor vibration characteristics.

The implementation of sensors has shown to be a viable option towards a solution for FDP mechanisms. However, limitations are inherent due to the invasive nature of wearable sensors, in addition to sparsity of available training data for niche sensor configurations. To this end, despite the comparability of input data with that of visual methods, sensor-based datasets offer insight and methodologies that are applicable in the construction of a fall dataset in spite of dissimilar fall data format.

C. AUDIO-BASED FDP SYSTEMS AND DATASETS

There are very few datasets and FDP systems that focus on audio-based fall detection. In [9], visual and audio fall detection methods were developed using a self-procured dataset inclusive of phonetic audio associated with a fall, in addition to visual fall data. The visual fall detection used a histogram-based classification of silhouette extraction and postural recognition, combined with an audible call for help. Similarly, [10] used visual data, combined with verbal, and environmental audio from egocentric cameras. The method was evaluated using decision fusion techniques for various visual learning models, and Mel-frequency Cepstral Coefficient audio features.

These techniques leverage fall associated audio for fall detection although verbal utterances accompanying a fall are a prevalent constituent in the analyses. Despite the distinguished advantage of audio in FDP techniques presented by these works, further investigation should address instances of falls resulting in, or causing unconsciousness, incoherence, or compromised speech [28]. These instances are in correlation with limitations inherent in audio FDP systems, and audio fall data availability. However, these further steps would require the development of a new fall dataset that includes the required audio data, since such a dataset is not publicly available.

Impact audio is an inherent constituent in falls of conscious and unconscious persons alike. An FDP designed to detect impact audio has the potential to recognize a fall in events where a person is unresponsive. The sparse nature of dynamic impact audio of falls prompts the development of a resource inclusive of such. Impact audio data would thus provide means for nuanced autonomous fall detection.

The presented work is aimed towards providing a resource effectively capturing diverse fall data enabling development of both visual and audio FDP techniques. The visual portion of this work is intended to provide a comprehensive

TABLE 1. Related datasets and their specifications.

Reference	No. of ADLs & (FEs)	Safety Equipment	Input Type	Recording Equipment*	No. of Participants
SisFall [21]	19 (15)	Landing mats	Sensor data	A, G	38
FallAIIID [22]	44 (35)	Wearable protection	Sensor data	A, G, M, B	15
MobiFall [23]	9 (4)	Landing mats	Sensor data	A, G	24
TST [24]	4 (4)	-	Image, Sensor	I, D, IMU	11
UMA [25]	8 (3)	Mattress	Sensors	A, G, M	17
MUVIM [26]	25 (undefined)	Mattress	Image	T, D, IR, I	40
CAUCAFall [27]	5 (5)	Wearable protection	Image	I	10
URFD [14]	5 (2)	-	Image, Sensor	D, I, IMU	5
CIRDO Corpus [9]	28 (5)	-	Image, Sound	A, I, D	17
EGOFALLS [10]	9 (4)	Mattress	Image, Sound	A, IR, I	14

*The key for column 5 is as follows: A: accelerometer, G: Gyroscope, M: Magnetometer, B: barometer, IMU: Inertial measurement unit, D: Depth sensor (most of these listed here use the Microsoft Kinect), T: FLIR Thermal image, IR: Infrared image, A: Audio, I: RGB image

collection of human falls recorded from various perspectives. Additionally, the audio component is meant to provide a multi-perspective collection of exclusively percussive audio fall data, not inclusive of verbal noise.

D. HUMAN FALL DATASETS

Human fall datasets (HFDs) are a significant contribution to the development of learning-based fall detection methods [29]. Fall datasets typically consist of orchestrated falls, which safely produce data that imitates characteristics of an actual fall. Typical constituents of HFDs include fall events (FEs), which exemplify fall characteristics, and activities of daily living (ADLs), which provide characteristics of regularly anticipated movement.

Established HFDs are listed in Table 1. There are commonalities shared between each of the listed datasets, although they present human fall data in different capacities. Each dataset interprets fall dynamics from distinct reference frames, capturing and quantifying signals produced by the motion. The presented datasets capture unique signals using various methodologies, and sensing equipment.

Datasets such as in CIRDO Corpus [9] and MUVIM [26] employ methodologies depicting ADLs and FEs for scenario-based recording. In contrast, isolated instance-based recording of FEs and ADLs separately is typical of the other datasets presented.

In [21], the SisFall dataset is presented, which is comprised of acceleration and rotation data using two accelerometers and a gyroscope from thirty-eight participants. The sensors were placed at the waist of each participant. Additional preliminary analyses were included to test the dataset, using fourteen extracted features and two threshold-based classifiers. Activities were performed in classrooms using a landing mat to ensure participant safety.

Similarly, [22] presents a motion signal dataset, referred to as FallAIIID, as a result of an accelerometer, gyroscope, magnetometer, and barometer readings placed at waist, wrist, and neck positions, for numerous movements performed by fifteen participants. Acceleration-based fall classification was conducted using deep learning and machine learning

techniques for the FallAIIID dataset. Participants performed movements using protective gear including a helmet, arm protection, knee pads, and a back protector, eliminating the need for a dampening fall pad to minimize data interference.

The MobiFall dataset in [23], presents smartphone accelerometer and gyroscope data collection for twenty-four participants. The dataset was recorded using smartphones providing data for two primary approaches to fall detection: threshold-based and machine learning methods. The smartphone was placed naturally in a pocket of the participant during recording of movements. Additionally, a landing mat was used to protect participants while performing FEs.

The TST Fall detection dataset presented in [24], provides RGB-D visual and acceleration data from two accelerometers. Classification of visually determined joint positions and accelerometer angle and acceleration from eleven volunteers were used to produce a dataset evaluating combined visual- and sensor-based fall detection. Analyses highlight the benefit of data synchronization between sensors. Data was recorded using a singular depth camera, in addition to two wrist and a waist mounted IMUs for eleven participants.

Using a combination of a smartphone, accelerometer, gyroscope, and magnetometer, the UMAFall dataset [25] provides sensor data from seventeen participants for five sensor locations. Comparative acceleration magnitudes for individual sensing points were presented for classification. Sensors were attached to the ankle, waist, right wrist, chest, and a smartphone was carried in the right pocket. Three unique environments were used to collect data, implementing mattresses to prevent participant injury while performing falls.

The MUVIM visual dataset was collected in [26], providing RGB, infrared, depth, and thermal image data. All cameras were mounted to the ceiling and were used to record forty participants. Visual data for each camera was comparatively evaluated using a 3D CNN auto-encoder. The study included scenarios that contained both scripted ADLs and FEs. Not all recordings were inclusive of both ADLs and FEs, some only simulated ADLs. Furthermore, the safety equipment used was a crash mat to cushion the fall of participants performing FEs.

The visual URFD dataset, in [14], presented RGB and depth data in combination with inertial data. Accelerometer, gyroscope, and single camera visual and depth data were collected for five participants undergoing threshold-based sensor classification, SVM, and KNN visual classification. This system effectively integrates both motion and visual data purposed towards a higher capacity to distinguish falls from daily activities. This study implemented a singular depth camera, and an IMU worn on a user's pelvic region. Five participants performed eight different activities, using a carpet to prevent injury during the falling movements.

The CIRDO Corpus dataset, in [9], recorded audio data using two microphones and visual data with three cameras, for seventeen participants. Cameras were placed on one side of the room facing the incident straight on, and the other at a forty-five degree angle and audio was captured using two microphones. One microphone was suspended from the ceiling, while the other was positioned on a nearby piece of furniture. The audio data included various forms of participant vocalization. Audio classification was achieved using an Automatic Speech Recognition (ASR) system, and visual data was classified using a histogram-based approach. Recorded activities captured ADLs and FEs in a shared scenario. A smart room environment was used, as it resembled a typical living space, and falls were performed on a carpet.

The visual and audio EGOFALLS dataset, presented in [10], used RGB and IR cameras to record fourteen participants. Visual classification was conducted using four learning models combined with features extracted from the audio using Mel-Frequency Cepstral Coefficients (MFCC). Cameras were placed egocentrically to collect RGB, infrared, and audio data. Classification schemes used late decision fusion, integrating descriptors from visual and auditory data. The audio data included ambient and verbal sounds surrounding the event. The egocentric camera positions were on the neck and waist of fourteen participants.

The BiMP dataset is uniquely intended to present multi-modal (visual and audio) representations of FEs and ADLs from a diverse set of viewing angles and perspectives. Additionally, the audio data provided in the BiMP dataset captures the fall impact noise independent of vocal interjection.

The ADLs and FEs used in the BiMP dataset were selected with reference to those used in previous datasets. Additionally, the modalities of the BiMP dataset are evaluated independent of one another to establish the fall characteristics inherent in each modality. The following methodology provides a comprehensive explanation of constituent attributes of the BiMP dataset.

III. METHODOLOGY

Effective learning-based fall detection relies on a dataset that accurately aligns with the intended application, is diverse in composition, and provides sufficient volume to allow for performance evaluations [30]. Thus, this dataset is purposed

towards a robust and cohesive presentation of data, reflective of multiple anticipated real-world scenarios.

Accurate recreation of an anticipated scenario demands consideration and minimization of variation between anticipated and recorded events. This section provides the background information, rationale, and the framework for the various aspects involved in the construction of the fall dataset.

To achieve a fall dataset comprised of accurate, multi-perspective visual and audio data, it is important to clearly formulate the required constituents defining these qualities. Therefore, the methodology for data collection must consider factors affecting the desired fall data. These factors include: the particular fall characteristics, the activities of the study and movement definitions, the environment in which the collection happens, participant demographics and characteristics, and the equipment used to record the data.

The following sections provide further insight around the factors listed above, beginning with the fall characteristics inherent in the target demographic of older adults.

A. FALL CHARACTERISTICS

Accurate data collection requires the integration of innate characteristics associated with falls of older adults in order to reproduce similar motions. The nature of falls in older individuals according to [31], concluded that the most prominent culprit responsible for falls was due to improper shift of the individual's weight. Furthermore, investigation in [32], indicated that longer choice step reaction time (CSRT) resulted in higher risk of falling. The relationship drawn between aging and longer CSRT implied older adults subject to falling would also be typically subject to a limited reaction time, thus, resulting in a fall where reactionary reflexes lag behind. This results in a fall where the impact is not cushioned by appendages or limbs in a manner that a person with a typical reaction time would be able to mitigate the severity of the impact associated with a fall. Similarly, [33] offered support for the idea that impaired reaction time placed older individuals at higher risk of injury in the event of a fall.

Not all older adults are subject to prolonged reaction times or higher impact severity. Some older adults are able to cushion their falls in a timely manner; however, it is not common. In [34], reaction times in arm responses between a younger demographic and an older demographic during a fall were observed. The findings of this study support the idea that reaction time is prolonged with an age increase. This paper also indicated that the reaction of older individuals, in terms of physical response to a fall, is to use their arms to dampen their fall.

Aside from a lower capacity for optimal reaction time, there is little indication that older adults are not inclined to reactionary measures of using their hands to cushion a fall, albeit with a delayed response time. The reflexive instinct for a person to use their hands and arms to dampen a fall transcends the age differential and remains a prevalent mode of fall mitigation among older adults.

This information offers an opportunity for increased accuracy in the reproduction of a fall. Delayed reaction time in the event of a fall was thereby incorporated into the data collection as part of the instruction for imitating falls of older adults. When falling, participants were asked to react naturally in their reflexes, though to delay the action. These considerations were instituted for the activities each participant performed.

B. ACTIVITIES OF THE STUDY

The human fall datasets listed in Table 1 collectively present an extensive range of daily living and fall movements. Nonetheless, many of the movements included in those datasets share underlying characteristics which can be grouped into fundamental movements shared across the datasets presented in Table 1.

Due to the subtle noise production of regular ADLs in comparison with the FEs, the incorporation of the miscellaneous ADLs offer various common household noises that differ from that of a fall in order to diversify the audible component of this dataset. These miscellaneous ADLs take the form of dropping an object or using an appliance typical of the given environment (e.g., pulling out a chair, opening/closing an appliance door, moving a laundry basket). As mentioned previously, each of the miscellaneous ADLs are listed in Table 3 and are environment specific.

TABLE 2. ADLs and FEs included in this dataset.

ADL Key	Movement	FE Key	Movement
ADL1	Standing	FE1	Forward
ADL2	Walking	FE2	Backward
ADL3	Jumping	FE3	Sideways
ADL4	Sitting down	FE4*	To a seated position
ADL5	Standing up	FE5*	From a seated position
ADL6	Squatting		
ADL7,8,(9)	Miscellaneous		

*Movement FE4 starts in a standing position and falls while descending to a seated position. Conversely, FE5 begins in a seated position and a fall occurs as a standing position is assumed.

TABLE 3. Miscellaneous ADLs unique to each room.

ADL Key	Movement	Room
ADL7	Pulling/pushing a chair	a
ADL8	Dropping a plate	
ADL9	Dropping a cup	
ADL7	Dropping a book	b
ADL8	Dropping a pill bottle	
ADL7	Knocking over a glass	c
ADL8	opening a microwave	
ADL7	Using the washer door	d
ADL8	Moving a laundry basket	
ADL9	Dropping a detergent container	e
ADL7	Dropping a book	
ADL8	Dropping a pill bottle	
ADL9	Closing a trunk	

The ADLs in this table provide additional data with higher variability and deviation from the common ADLs incorporated consistently throughout all environments.

An explanation on how to perform movements listed in Table 2 was given to each participant prior to recording. The recording of a movement would then begin once the participant assumed the initial standing or sitting position that would be proceeded by the movement (i.e., a participant would initially be in a seated position for ADL5 and FE5). Once recording began, each participant would silently count to ten and then perform the movement. Once the participant had completed the movement, the recording would stop three seconds after all motion stopped. The environments where the activities were performed are further explained in the following subsection.

C. ENVIRONMENT

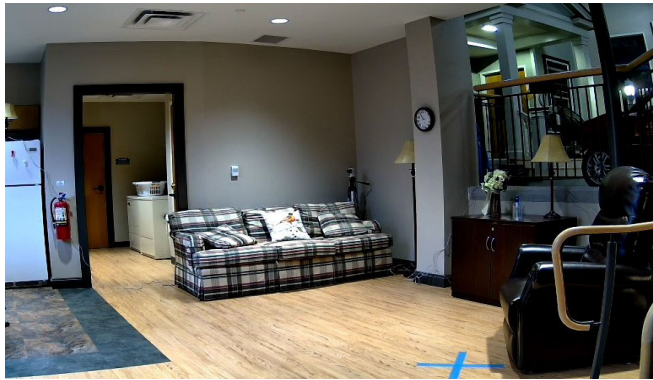
The previous section addressed procedural variables associated with bodily motion, while this section addresses the environmental considerations.

Background information can significantly influence learning networks as it can affect attributes such as training bias, context, and feature extraction. Both [35] and [36] present examples of how image background impacts the ability of the learning network. Although these works are not directly concerned with the development of human fall datasets, they communicate the influence background information can have on learning networks. The significance of image background in the context of learning, demonstrates the importance of data collection in an environment that conveys the visual ambience of the intended application.

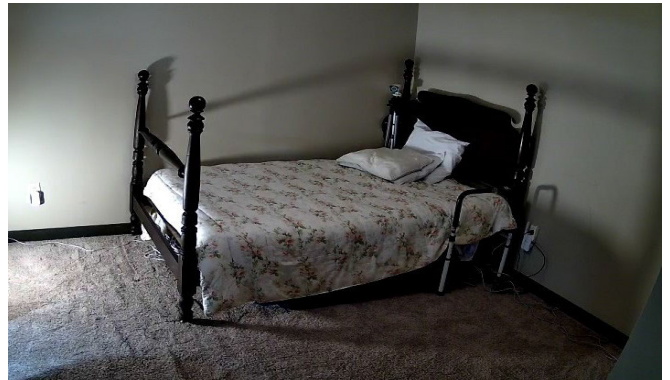
Visual background information included in this dataset was captured in a mock home laboratory. The setting was chosen for the relevance it provides within the context of this work, as it replicates an environment where older adults are statistically prone to fall [3].

The laboratory rooms used in the data collection are: (a) living space, (b) bedroom, (c) kitchen/dining room, (d) laundry room, and (e) secondary bedroom. Due to the relatively low percentage of falls that actually occur within the bathroom [37], and privacy implications, a bathroom setting was not included in this data collection. Additionally, outdoor environments have been omitted from data collection as this work is aimed towards further development of aging in place technologies. Concern is therefore prioritized within the home rather than the external environment. It is worth noting that in addition to privacy challenges related to the placement of fixed cameras in bathrooms, bathrooms are often constrained spaces with unique elements dissimilar to the remainder of the home (e.g., obscuring curtains, smooth tubs, wet features, locking doors), which can make it difficult for mobile robots to access or observe. Similarly, fixed cameras are typically impractical in most outdoor environments, and outdoor environments can have challenging geographic features, poor lighting conditions, and inclement weather that restrict the use of mobile robots for data capture.

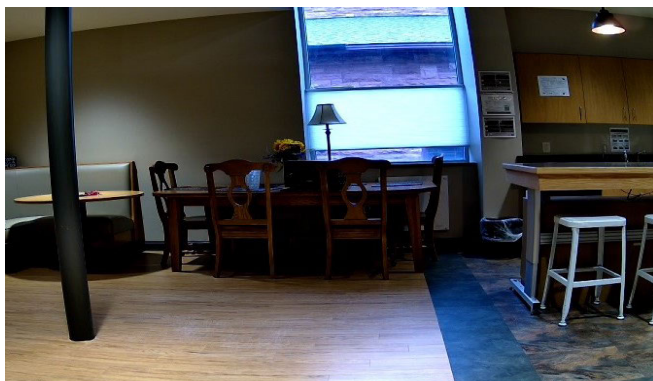
The rooms depicted in Fig. 1, where data collection was performed, are Clarkson University's Simulation Laboratory.



Room (a): Living space



Room (b): Bedroom



Room (c): Kitchen/dining room



Room (d): Laundry room Room (e): Secondary bedroom

FIGURE 1. Views of each rooms: (a) living space, (b) bedroom, (c) kitchen/dining room, (d) laundry room, (e) secondary bedroom room.**TABLE 4.** Participant characteristics overview.

Characteristic	Mode	Mean	Median	Min	Max	SDI
Age	24	27.6	24	18	57	0.8704
Weight (kg)	70	81.28	78	53	104	0.9376
Height (m)	1.63	1.74	1.77	1.47	1.91	0.9472

This laboratory is structured as an imitation home purposed towards facilitating realistic home care. The appearance of the laboratory offered facilities that aligned well with the environmental requirements of this dataset. The inclusion of these rooms offered diverse backgrounds similar to what is expected in a residential setting. In addition to the environmental constituents of the dataset, participant attributes are provided in the following section.

D. PARTICIPANT DEMOGRAPHICS

All participants involved in this study were voluntary with the option to opt out of participation at any given point of completion. Exclusion criteria for participation consisted of individuals younger than eighteen years of age, and any participants allergic to the materials of the safety equipment. A total of twenty-five volunteers participated in performing FEs and ADLs for this study. Participant characteristics such as: age, weight, and height, are summarized in Table 4 in

TABLE 5. Participant demographic overview.

Demographic Grouping		Population	Percentage (%)
Gender	Female	6	24
	Male	18	72
	Trans Female	1	4
Ethnicity	Asian	2	8
	Black	1	4
	White	21	84
	Hispanic/Latino	1	4
Dominant hand	Left	3	12
	Right	22	88

terms of the statistical distribution. Furthermore, Table 5, presents the dataset composition in terms of participant gender, ethnicity, and dominant hand. Participants were not required to disclose their gender identity on the preliminary questionnaire and did so of their own volition.

Among the statistical quantities included in Table 4 is the Simpson's Diversity Index (SDI) for each characteristic calculated using the definition provided in [38]. The SDI computes the likelihood of two random samples belonging to the same category. The SDI values range from zero (no diversity) to one (infinite diversity). As can be seen in Table 4, the physical participant characteristics have relatively high diversity. The SDI was not computed for the demographics

in Table 5, since by definition the diversity values would be low due to the limited discrete number of groupings (e.g., two groupings for dominant hand). If desired, these values can also be computed directly given the values in Table 5. Further discussion on the limitations of the participant demographic diversity can be found in Section VI-B.

The exclusion criteria of age and allergy considerations did not affect any volunteers that signed up to participate in the study resulting in the involvement of all volunteers. The voluntary participation and study location, therefore, were the only influences on the demographic and characteristics of study participants.

Each participant performed ADLs and FEs in a specific room. Four laboratory rooms (a)-(d) were used to record six participants each. None of the participants appear in multiple rooms, meaning no two scenarios feature the same participant. A single participant was recorded in the remaining room, i.e., room (e). Following are specifications of the equipment used while recording the data.

E. EQUIPMENT

Data was collected with a Reolink RLK8-800B4 system [39], comprised of four 3840p×2160p, 20 frames per second cameras, combined with a two terabyte network video recorder. Cameras were placed as evenly around the room as possible, each of which viewed the same central location.

The positioning of the cameras in this fashion offered multiple viewing angles of the same incident, while also recording audio data from diverse spatial regions. In each room, cameras were placed as evenly as possible around the incident to avoid data similarity. The vertical placement of each camera in this study ranged from 90 cm to 180 cm. While not specifically designed for mobile humanoid robots, this range of heights is well-suited for the training and testing of FDP techniques in mobile robotic systems [40].

The equipment used for safety included a vest for torso padding, a helmet, knee and elbow pads, and gloves. Safety equipment used for the fall events was worn as shown in Fig. 2. The knee and elbow pads were equipped with adhesive cloth to imitate clothing. The pads were covered in cloth in order to provide audio mimicking the impact sound of a clothed joint opposed to that of a safety pad.

Each participant wore this safety equipment while performing all FEs. Similar to the unwanted interference landing mats play in accelerometer data collection [22], audio data collection is subject to the same influence. To avoid any potential interference from the landing mat on the integrity of data collection, this study opted to use wearable safety equipment.

Comprehensive inclusion of the the spatial, movement, and participant attributes provide context for the recorded data, which was then processed and repositied for further analysis.



FIGURE 2. Safety equipment used: 1) helmet, 2) vest, 3) elbow pads, 4) gloves, and 5) knee pads.

IV. DATA FORMATTING AND STORAGE

After recording each participant, data was moved to a folder in a secure, password protected server. A folder was assigned to each room labeled according to rooms (a)-(e). Within that folder, additional sub-folders, were assigned for each participant, labeled by the letter of the room wherein the recording was taken, followed by the number corresponding to the order in which that person participated, referred to as the *participant identifier*. Each participant folder includes two sub-folders labeled *ADL* and *FE*. Inside these folders are the video recordings of each participant.

Videos were stored as .mp4 files, and were labeled according to the movement keys listed in Tables 2-3, being preceded by their participant identifier. The organization of folders and stored data follows the directory structure presented in Fig. 3. The figure displays folder hierarchy and visual fall data. The ADL data was identically formatted and stored. The audio data were stored as .wav files using the same folder structure as the visual data.

All identifiable personal attributes have been removed from the collected data, meaning all faces have been blurred, all tattoos have been blurred, and there are no names associated with any participants in the presented data. The demographic and characteristic information for each participant was labeled and stored with the collected data. Participant information was labeled in accordance with their participant identifier. Formatting data involved rotating, blurring, and trimming videos, and extracting audio files.

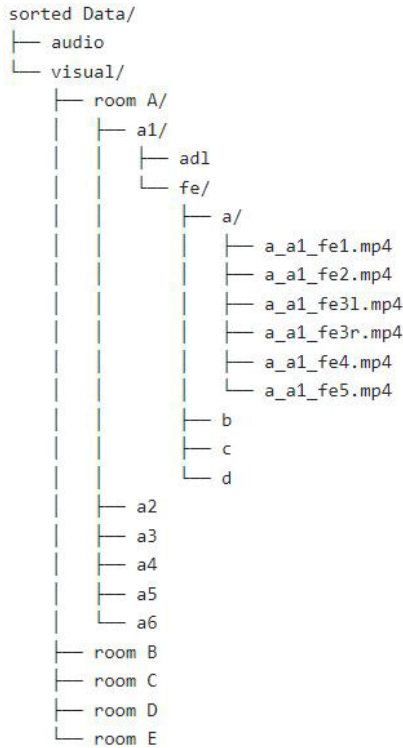


FIGURE 3. Organization of the stored data and hierarchy of the folders.

Personally identifiable information was removed from the dataset, which can be made freely available in compliance with the limits of the IRB approval.

V. ANALYSIS

This analysis serves to evaluate both the visual and audio dataset components for fall detection. Both audio and visual constituents were independently evaluated using unique and distinguishable class features. Thus, classification techniques were employed for audio and visual data independently in order to quantify the recognizable fall characteristics inherent in each modality. The dataset was used to train and test classifiers for both data modalities, purposed towards binary classification of FEs as falls and ADLs as nonfalls.

This work prioritized the classification of falls over specific movement classifications, thereby removing the necessity of a non-binary classification scheme. More specific movement classification beyond fall or nonfall is beyond the scope of this research, but is achievable using this dataset.

TABLE 6. Classifier input and data preparation.

Classifier	Label	Input	Data Preparation
GoogLeNet	V1	Visual	Frame separation and merging
GoogLeNet-LSTM	V2	Visual	Frame separation and sequential frame analyses
GoogLeNet	A1	Audio	Wavelet transform heatmap
GoogLeNet	A2	Audio	Spectrogram heatmap

A. EVALUATION

Classification of the dataset was conducted using multiple methods in order to establish its viability for different analyses. Each method of classification was supported by two sets of validation metrics and involved training neural networks. In the context of this work, neural networks constitute a classifier and are able to identify patterns and similarities in data. Each classifier was initially trained and tested using a 70-30 data split using data from rooms (a)-(d). The remaining data from room (e) was then used as secondary test data for the trained model. The training set included of 889 sequences and 381 complementary test sequences. The secondary test set from room (e) included 56 files.

Beyond the data partition, both visual and audio portions of the dataset have each undergone two unique classifications. Each classifier for both modes of data input and the associated specifications are listed in Table 6. Each of the classifiers were trained on the dataset, and validated on it twice.

The model used for each analysis of both the audio and visual data is a GoogLeNet architecture [41] consisting of 144 layers that contribute to 11 major modules. The common model between the analyses is presented in Fig. 4, depicting a GoogLeNet architecture.

Using the GoogLeNet, an input image undergoes multiple layers of convolution and max pooling prior to the inception modules presented in Fig. 4. In total, the model uses nine inception modules comprised of convolutional, pooling, and depth concatenation layers. The light blue represents convolutional layers, green represents max pooling layers, orange represents depth concatenation layers, purple represents a

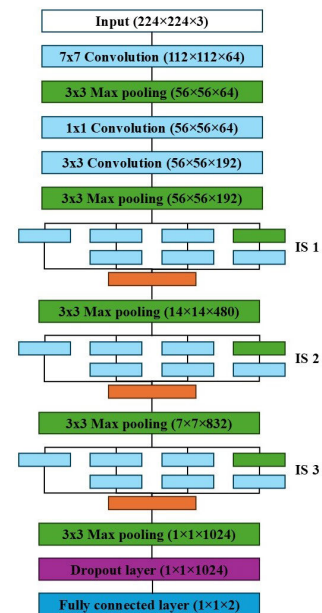


FIGURE 4. High level GoogLeNet architecture diagram. Colors associated with layer type: light blue-convolutional, green-max pooling, orange-depth concatenation, purple-dropout, dark blue-fully connected.

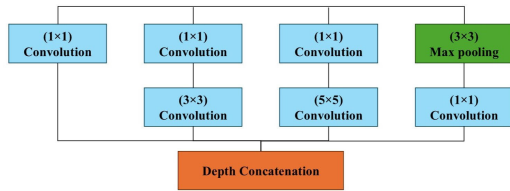


FIGURE 5. GoogLeNet inception module layer breakdown, with the same color scheme used as in Fig. 4.

dropout layer, and dark blue represents a fully connected layer.

GoogLeNet was used for its versatility in capturing spatial features of different sizes. The ability of the GoogLeNet to capture both local and global spatial dependencies, offers high adaptability for the varying input formats and data. The applicability of a GoogLeNet architecture thus accommodates the data preparation techniques described in Table 6. The diagram shows the structure of the inception modules, and the number on the right of each module indicates how many of them are concatenated between max pooling layers. The nine inception modules make up three inception sequences, denoted by the number to the right of the inception modules in Fig. 4.

The architecture of the inception modules leverages different convolutions in parallel, capturing features of various dimensions. The layers and architecture for the inception modules are shown in Fig. 5.

Each inception module provides multi-level feature extraction, where medium-sized spatial features are captured by the (3×3) convolutions and larger spatial features are captured by the (5×5) convolutions. These inception modules are sequentially connected in between max pooling layers, creating inception sequences (IS) for further feature extraction.

The inception sequence of tandem inception modules, as shown in Fig. 4, are labeled according to the number appearing next to the module. The sequences are structured as follows:

- IS 1: 2 sequential inception modules
- IS 2: 5 sequential inception modules
- IS 3: 2 sequential inception modules

Each inception sequence captures information at different levels as a result of combined outputs of elements from the processing layers of the module. The first inception sequence extracts basic low-level features. The initial sequence focuses on finer details in the feature extraction, such as edges colors and textures. The second inception sequence is responsible for intermediate features, capturing more context than the preceding sequence, such as shapes or objects. The subsequent and final inception sequence captures the highest-level, most abstract features, representing patterns or objects of higher complexity.

The GoogLeNet architecture is a common thread in all visual and audio data evaluations presented in this study. The

TABLE 7. Classifier hyperparameters.

Classifier	Batch Size	Epochs	Learn rate		
			Initial	Drop Period	Drop Factor
V1	60	5	5×10^{-5}	-	-
V2	20	6	5×10^{-5}	1	0.5
A1	18	8	1×10^{-4}	-	-
A2	18	8	1×10^{-4}	-	-

same structure was used for all evaluations, with the addition of a long short-term memory (LSTM) network for a portion of the visual data evaluation. To preserve temporal information, an LSTM is appended after the GoogLeNet CNN. The addition of the LSTM incorporates features extracted from individual frames as part of an evaluation across the entire sequence. This evaluation scenario modifies the structure of the model described in Fig. 4 by inserting a bi-directional LSTM intended for capturing temporal features of the visual data, between the final max pooling, and dropout layer.

Training was completed using a Intel(R) Core (TM) i7-7700 CPU @ 3.60 GHz processor and 64.0 GBs of RAM. Training times would take up to two days to complete, using hyperparameters listed in Table 7. The final tuned hyperparameters were the product of iterative trials and adjustments.

Prior to classification, data was formatted to match the classifier (length $\times$ width $\times$ RGB color channel) input layer of the size 224 pixels $\times$ 224 pixels $\times$ 3. The model was designed for image classification, thus, resulting in each individual recording being rearranged to align with the model architecture.

The conversion of audio signal to a pictorial representation was accomplished through two processing techniques: wavelets (Section V-C1) and spectrograms (Section V-C2). These two signal processing techniques render a heatmap image representing the audio signal associated with each recording.

B. VISUAL COMPONENT CLASSIFIERS

The two approaches used to classify visual data involve a GoogLeNet single image classification, and a GoogLeNet-LSTM sequential image classification. Aside from the resizing of video frames to match the model input requirements, image augmentation was not applied in this analysis but may be considered for future work. The required preparation and handling of data are listed in the third column of Table 6.

The GoogLeNet without the LSTM was used to classify the images that were a result of merging frames from a singular video into a single frame. Each video was split into individual frames that were resized to (224 pixels $\times$ 224 pixels) and stored in an array with the corresponding RGB values. Frame sequences were represented by the average RGB value across all frames for each pixel location, which produced a singular averaged image for classification.

The addition of the LSTM involves temporal aspects of the data in classification. Converting a video for this model involved frame extraction, resizing, and sequential storage. Similar to the merged array, this array was structured with the same pixel size (224 pixels $\times$ 224 pixels) for both height and width, where each pixel in a frame maintained the same RGB values.

The GoogLeNet diagram in Fig. 4 depicts the model used for feature extraction for both analyses, along with classification of merged images. The GoogLeNet-LSTM modifies the network presented by redirecting the GoogLeNet global pooling output into a sequential feature array, passing it to an additional sequence input layer. The input layer takes each sequence element as a feature vector. The sequences are processed twofold by a bi-directional LSTM layer, followed by regularization and the conversion of LSTM outputs to probability space outputs for classification. The LSTM introduced additional complexities with the inclusion of temporal features, thus increasing the volume of learnables. The additional complexities were mitigated by implementing a learning rate drop for more precise weight adjustments over time.

C. AUDIO COMPONENT CLASSIFIERS

The classification of the audio portion of this dataset used the same GoogLeNet architecture displayed in Fig. 4. The GoogLeNet architecture accepts images for classification, thus, audio data was formatted as pictorial inputs. Raw audio signals were extracted from each video and were processed without filtering. The exploration of filtering processes extends beyond the focus of this work and provides a basis for future analyses that may prove to be advantageous regarding data clarity or classification performance. The experiments were conducted in closed environments focused on the noise generated by the activity. It was expected that sounds generated by the participants performing the activities would contribute to the background noise. However, separation of mixed audio signals produced by concurrent sources is not a trivial task [42], and deviated beyond the scope of this work. Depending on the intended application, the addition or removal of background noise from the dataset may be necessary to improve fall detection outcomes. The noise present in the current dataset can be considered reflective of a quiet home environment. Similarly, filtering techniques may be considered in future work as none have been included as part of the analyses in this work.

Audio signals were converted into images through the application of wavelet and spectrogram transforms, capturing the temporal data within a single image. The transforms return the frequency and amplitude constituents of an audio signal over time. Frequency quantifies the amount of repeated vibrations captured in a signal over a given period of time. The amplitude of a signal, is the magnitude of the vibration experienced at a given time or frequency. Short-time Fourier transforms and continuous wavelet transforms were chosen

for their unique abilities to produce time-frequency detailed representations of signal characteristics as a learnable image [43].

1) WAVELETS

Each wavelet heatmap was the result of a continuous wavelet transform (CWT) applied to the audio signal associated with the recorded movement. The wavelets provided time-frequency representations of the audio signal. The CWT of continuous audio signal from a fall event with a wavelet function is defined in (1) [44]:

$$CWT(s, \tau) = \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} x(t) \hat{\psi} \left(\frac{t - \tau}{s} \right) dt \quad (1)$$

where the associated wavelet coefficients are the scale factor, s , the translation factor, τ , the complex conjugate of the wavelet function, $\hat{\psi}(t)$, and the continuous audio signal, $x(t)$.

The bump wavelet defined in (2) [44] is the wavelet function used in the CWT and it maintains spectral content of signals over time.

$$\hat{\psi}(\omega) = \exp \left(1 - \frac{1}{1 - \left(\frac{\omega - \omega_{peak}}{\sigma} \right)^2} \right) \quad \text{for } |\omega - \omega_{peak}| < \sigma \quad (2)$$

The frequencies, ω , at a given time, t , center the wavelet around the peak frequency, ω_{peak} , for an effective wavelet range defined by bandwidth, σ .

The scale factor, s , refers to the compression or enlargement of the wavelet. Higher values for the scale factors correspond to lower frequencies, and lower values are associated with the higher frequencies. Coarse signal features are dependent on a larger scale value; whereas, smoother features are dependent on a smaller value.

The translation factor, τ , refers to the incremental shift of the wavelet for a given time step. These wavelet coefficients are obtained after applying the wavelet to a signal, and are representative of the wavelet function compared to the continuous signal.

A pictorial representation of the CWT can be seen in Fig. 6(a) and 6(b) in the form of a heatmap. Fig. 6(a) is the wavelet transform of the audio from an ADL, and Fig. 6(b) is of an FE. The wavelet transform heatmaps represent the scale as frequency along the y-axis for a given time on the x-axis. Both subfigures are accompanied by a scale (i.e., the color bar right of the subfigures) corresponding to the magnitude of the wavelet coefficients. Higher magnitudes indicate a stronger correlation of the wavelet with the signal for a given scale and translation value. The benefit of the transform in this form is that it presents transient features, or sudden changes in the audio signal as visibly identifiable characteristics.

2) SPECTROGRAMS

In addition to the CWT, spectrograms of short-time Fourier transforms (STFT) facilitate similar visual reconstruction of a

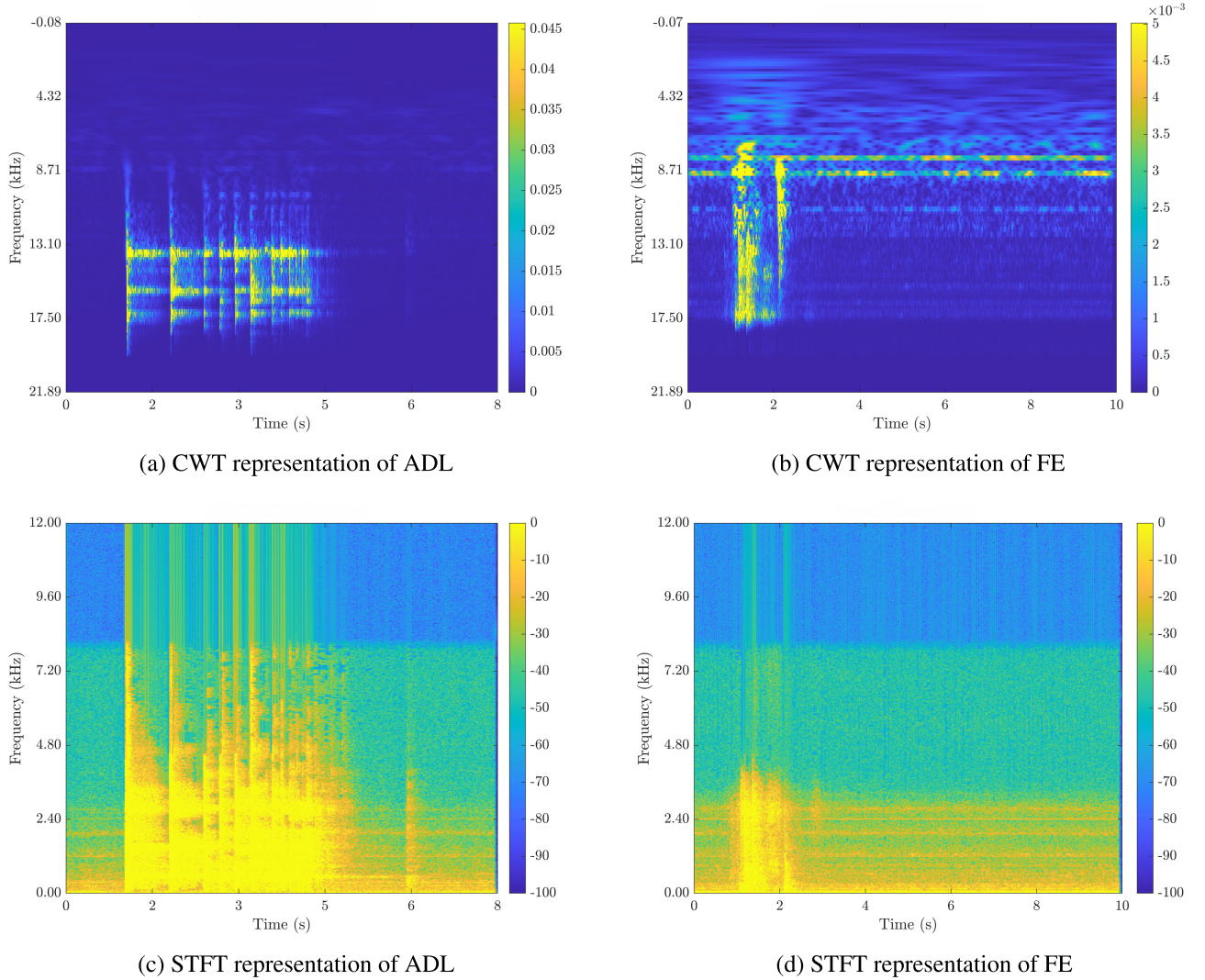


FIGURE 6. Comparison of example continuous wavelet and short-time Fourier transform heatmaps for ADL and FE conditions.

signal. The STFT is a signal processing technique for analysis of a signal where the frequency profile changes over time. This provides representation of a signal in both the time and frequency domains.

The process applies the Fourier transform to short overlapping signal segments. The result is a time-frequency representation of the frequency profile for the signals duration. The definition of the STFT is presented in (3) [45]:

$$\text{STFT}\{x(t)\}(T, \omega) \equiv X(T, \omega)$$

$$X(T, \omega) = \int_{-\infty}^{\infty} x(t)w(t-T)e^{-i\omega t} dt \quad (3)$$

where $x(t)$ is the input signal, or the audio signal in this case, $w(t-T)$ is the window function centered at a given time, T , and $e^{-i\omega t}$ is the complex exponential representing the Fourier transform.

The window function $w(t-T)$ is used to isolate a segment of the signal $x(t)$ for analysis. The window function used is

the Hamming window and is defined in (4) [45], representing the window of observation centered at time T , and N being the number of discretized time segments of the signal.

$$w(t-T) = 0.54 - 0.46 \cos\left(\frac{2\pi(t-T)}{N-1}\right) \quad (4)$$

This function limits the contribution of the signal outside close proximity of time T . The temporal information is maintained through window overlap, as the window is incrementally shifted for each specified time.

The window function has piecewise features resulting in a non-zero value over a limited time duration, and zero elsewhere. The value for the function converges to zero at the duration boundaries. The analyses completed with the STFT utilized a window size of 50 ms with an overlap length of 25 ms. The window and overlap lengths produce an isolated signal segment. The isolated signal segments were

chronologically combined providing a temporal depiction of adjoined signal segments.

Time-frequency representations of STFT applied to audio signals are displayed in Fig. 6(c) and 6(d) for ADL audio and FE audio transforms, respectively. The figures exhibit a time-frequency representation of audio signals using STFT, where the y-axes represent frequency magnitude and the x-axes represent time. The color bar on the right of each subfigure represents the magnitude of the STFT in decibels.

The frequency range in the STFT spectrograms is a result of the positive frequency components of the complex conjugates of the negative frequencies. The magnitude spectrum is symmetric about a zero frequency reference, resulting in a spectrogram frequency range with a minimum constraint of zero. Signal processing of the audio signal in this fashion is an alternative to that of a wavelet, providing an alternate means of audio signal visualization.

D. CLASSIFICATION METRICS

The evaluation metrics for classification of both the audio and visual analysis were accuracy, recall, and precision. These metrics were used to evaluate the classification of the 70-30 test data, and the room (e) test data.

The measures of accuracy, recall, and precision are defined in (5), (6), and (7), respectively. These metrics are a quantification of the correct and incorrect classifications of the model. Each of these metrics are used to establish the integrity of the data presented and the viability of the dataset for the purpose of fall detection.

Each of these quantities are calculated using the number of correct and incorrect predictions. In accordance with the binary classification scheme, correct and incorrect predictions are indicated by *true* and *false* labels, respectively. Additionally, falls and nonfalls are represented by *positive* and *negative* labels, respectively.

Therefore, classification predicts the correct classifications as either a true positive (TP) or a true negative (TN), and incorrect classifications as false positives (FP) and false negatives (FN). Correct and incorrect classifications thus provide the variable for the evaluation metrics defined as:

$$\text{Accuracy (\%)} = 100 \left(\frac{TP + TN}{TP + TN + FP + FN} \right) \quad (5)$$

$$\text{Recall (\%)} = 100 \left(\frac{TP}{TP + FN} \right) \quad (6)$$

$$\text{Precision (\%)} = 100 \left(\frac{TP}{TP + FP} \right) \quad (7)$$

The accuracy quantifies the overall correctness of the classifications; whereas, the recall quantifies the ability of the classifier to capture all positive instances. Alternatively, precision measures the correctness of positive classifications.

Following are the results of applying the processes outlined within the context of this analysis.

TABLE 8. Visual classification metrics.

	Model	Accuracy	Precision	Recall
Training Split	GoogLeNet	94.87	97.49	92.82
	GoogLeNet-LSTM	96.15	93.33	99.31
Validation Set	GoogLeNet	91.07	96.55	87.50
	GoogLeNet-LSTM	94.64	100.00	90.63

TABLE 9. Visual positive and negative predictions.

	Model	TP	TN	FP	FN
Training Split	GoogLeNet	194	176	15	5
	GoogLeNet-LSTM	196	179	14	1
Validation Set	GoogLeNet	28	23	1	4
	GoogLeNet-LSTM	29	24	0	3

VI. RESULTS

This section presents a comprehensive evaluation of the BiMP dataset, employing the analytical techniques outlined in Section V. The primary focus is on the performance of the fall detection classification schemes and the ability of the dataset to portray relevant fall characteristics. Additionally, the dataset limitations are identified and discussed.

The results of using a GoogLeNet to classify merged frames of a video indicate a differentiable trend in fallen and nonfallen data formatted as a singular image. Results yielded by the LSTM in combination with the GoogLeNet likewise showcase relevant sequential characteristics of the intended motion as a recognizable aspect of the dataset. Complementary to the metrics displayed in Table 8, are quantifiable true and false classifications of the visual data in Table 9.

Similar to the visual evaluation of the dataset, the audio portion was also evaluated. Classification using CWT and STFT signal processing techniques display promising results, indicating adequate fall representation. Audio evaluation metrics are displayed in Table 10, with the associated true and false predictions in Table 11.

The first two rows of each of the Tables presented in this section display the results for the data collected in rooms (a)-(d) using a 70-30 training-testing split. These rows display the evaluation metrics during model training. The last two rows of each table include the results for data collected in room (e), which was used as a validation set.

A. CLASSIFIER EVALUATION

The scope of this work is focused towards creating a dataset for human fall detection; however, the validation of the dataset is established through the development of fall classification techniques. The developed fall detection techniques indicate the presence of clear human fall characteristics within the dataset.

The FN quantities from Tables 9 and 11 indicate the GoogLeNet-LSTM classifier provides the fewest misclassified falls in training, and fewer misclassifications than the others. Additionally, classification using CWT provided the

TABLE 10. Audio classification metrics.

	Model	Accuracy	Precision	Recall
Training Split	CWT	96.41	95.93	97.70
	STFT	95.64	94.62	97.69
Validation Set	CWT	92.857	100	87.50
	STFT	89.29	90.63	90.63

TABLE 11. Audio positives and negative predictions.

	Model	TP	TN	FP	FN
Training Split	CWT	212	164	9	5
	STFT	211	162	12	5
Validation Set	CWT	28	24	0	4
	STFT	29	21	3	3

second lowest amount of misclassifications of the validation set, although they are all misclassified falls.

Nonetheless, the accuracies of the classifiers used to detect falls indicate high capability of differentiating between ADLs and FEs. The STFT classifier performed less desirably with the validation set; however, it still demonstrates its viability for fall classification. The lower prediction accuracy may be resolvable in the future using a different neural network, or different STFT parameters.

The BiMP dataset offers visual and audio fall data for human fall detection, substantiated by the distinguishable characteristics inherent in the data. The percussive audio data associated with a fall impact promotes the development of fall detection techniques from proximate listening devices.

The metrics discussed in Section V-D establish a baseline to gauge the classifiability of the presented data. The evaluation of these metrics, presented in Tables 8 and 10 substantiate the classifiability of the dataset. The resulting evaluation of the dataset is indicative that this dataset has the potential to be a valuable resource for developing future fall detection mechanisms.

B. LIMITATIONS

The limited diversity of participant characteristics in the dataset is a current limitation. While participant demographics include various characteristics, the participants in the study should ideally include older adults exclusively. However, this is a known challenge due to the inherent safety issues involved in older adult simulation of fall events. Similarly, gender and ethnicity should ideally reflect a comparative composition of the senior population; whereas, the study demographics more closely reflect the geographic region wherein the dataset was developed, namely, rural, upstate New York, USA. In this dataset, the geographical limitations inherent to the area wherein this study was performed, namely, rural upstate New York, bias the demographics to those of the local region. The application of this dataset should consider both the specific target demographic, and for widespread implementations, the broader population diversity and characteristics. However,

demographic inaccuracies are inherent for any instance of training data noninclusive of the anticipated user(s) exclusively. Future work could consider collecting data from constant observation of older adults in their living settings to increase the fidelity of the dataset. However, aside from increased cost, observation duration, and privacy concerns, this also creates challenges related to both capturing multiple perspectives and variety in the characteristics of the falls observed.

The impact of dissimilarities present in the BiMP dataset participant and intended user demographics are not easily quantifiable, but have the capacity to influence classification metrics of various system architectures. The impact of demographic diversity on the widespread applicability of the models used is beyond the scope of this work. The models used in this analysis aid in the evaluation of the ability of the dataset to provide data detectable by various methods. The models are not intended for widespread replication and implementation, but rather to serve as a benchmark for the contents of the dataset.

Emphasis has been placed on the background used for data collection. While environmental variables are intended to replicate those characteristic of a home, it is impossible to account for all the application environments.

The safety equipment used was not typical attire of older adults prone to falling. The inclusion of safety equipment negatively affects the similarity of collected data and unanticipated falls. The safety equipment is a necessary precaution; however, its implementation produces the potential for unwanted elements affecting signal frequency of audio data, and visually inaccurate attire. Nonetheless, participant safety is paramount to any other aspect of this work.

Limitations are also inherent in the inability for orchestrated falls to perfectly display characteristics of a natural fall. Orchestrated and natural falls display different reactionary elements. The reflex response in each of these cases exemplify dissimilar body weight shifts and response times. These differences potentially influence visual data, and the impact responsible for the audio signal; nonetheless, minimization of these concerns was achieved by instructing participants to slightly delay their reflexes while falling.

VII. CONCLUSION

This dataset presents multiple perspectives of a singular incident, providing unrepeated activity recordings. Further, multi-perspective recording captures unique environmental fall audio characteristics, providing a basis for sound-based fall detection from varying listening locations.

Data processing using GoogLeNet singularly and with an additional LSTM showed promising results as both systems displayed aptitude for differentiating between falls and non-falls with accuracies ranging from 91%-96%. Much like the results from the visual analysis, CWT and STFT processing techniques used for audio fall detection exhibited accuracies ranging between 89% and 96% ability in distinguishing fallen and nonfallen states.

A foreseeable utility of this dataset leverages the prevalence of virtual assistants in homes. Virtual assistants provide new potential avenues for audio fall detection using smart home devices. The existent listening nature of virtual assistants and smart home devices within households provide a secure means for audio monitoring and therefore fall detection that deviates from privacy concerns associated with continuous visual monitoring. This concept has the potential for fall detection regardless of verbal instruction to the virtual assistant; this being important in the event of a fall resulting in unconsciousness. However, applications of this dataset are not restricted to a stationary device, as there is the potential for integration of FDP systems into mobile devices or robots equipped with microphones. Despite the promising results when evaluating both fall data information input formats, the potential for fall detection using both inputs simultaneously merits consideration in future work.

Potential for the combined use of input modalities include a fall detected by audio, which is succeeded by a form of confirmation using visual data during or after the incident; having applications in environments with any combination of listening or visual monitoring capabilities, including smart homes. Additionally the efficacy of weighted voting is being explored when using both input modalities. Further learning methods could incorporate aggregation or decision level fusion of individual classifiers.

Considerations for potential challenges regarding signal interference from additional sources are beyond the scope of this work. The reasoning behind these exclusions stems from the focus of this work to provide a basis to be built upon, meaning to capture the essence of the incident without it being convoluted with interfering signals. Thus, providing adequate basis for scenarios noninclusive of outside interference while simultaneously allowing the ability to add interferences specific to the objectives of further work.

The novelty of this dataset is highlighted in the presentation of variable perspective, multi-modal (e.g., visual and audio) fall data. Broad analyses leveraging image processing, signal transform, and machine learning techniques provide context for the efficacy of the proposed dataset. Promising classification results indicate distinguishable fall characteristics inherent in the dataset. Furthermore, these constituent fall characteristics are exemplified in their identifiability across diverse methods of processing and classification.

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